Reproducible Cross-Subject EMG Gesture Classification on NinaPro DB3:

Benchmark Evaluation, Domain Adaptation, Confidence Calibration, and Agentic Search

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Abstract

Cross-subject electromyography (EMG) gesture classification remains difficult because signal distributions vary substantially across users, creating a domain shift problem that limits reliable prosthetic control. This capstone builds a reproducible Leave-One-Subject-Out (LOSO) benchmark on NinaPro DB3 (11 transradial amputees, 17 gesture classes) and comprehensively evaluates eight methods under identical conditions. A controlled ablation experiment demonstrates that the large gap between our results and published baselines (Niu et al. 93.82%, Sandoval-Espino et al. 90.23%) is not attributable to subsampling: removing the training cap increases within-subject accuracy from 9.3% to only 10.7%, leaving a 79.5% gap explained almost entirely by evaluation protocol. Maximum Mean Discrepancy (MMD) analysis confirms that subjects S3 and S7 have the highest inter-subject distribution shift (MMD=0.596), and the Latent CNN shows a negative correlation between MMD and accuracy (r=-0.479), with low-MMD subjects achieving nearly double the accuracy of high-MMD subjects. Our novel cycle-consistent cross-subject autoencoder, trained with formally defined reconstruction, cycle-consistency, and classification losses, achieved the best LOSO balanced accuracy of 0.0755. An agentic search across 81 configurations found the optimal adversarial setup. Temperature scaling reduced ECE by 81% (MLP) and 91% (Latent CNN, ECE=0.0147). All findings converge: cross-subject EMG classification is highly sensitive to subject distribution alignment, and naive reproduction without LOSO evaluation leads to near-random performance under clinically realistic conditions.

Keywords: *surface electromyography, LOSO, domain shift, MMD, cycle consistency, adversarial training, confidence calibration, agentic search, NinaPro*

1. Introduction

Surface electromyography (sEMG) pattern recognition enables amputees to control prosthetic limbs through muscle activity [1]. Published accuracy in the sEMG literature varies dramatically with evaluation protocol: within-subject methods report 70-90% while cross-subject (leave-one-subject-out) evaluation reveals near-chance performance for the same models [4,5]. This discrepancy is caused primarily by evaluation protocol design, not model capability.

This capstone provides the first systematic quantification of this gap on NinaPro DB3 through controlled ablation, introduces a novel cycle-consistent autoencoder with formal multi-component loss, and establishes Maximum Mean Discrepancy as a proper domain shift metric for this dataset. Beyond accuracy, model confidence scores in published work are uncalibrated, assigning 90% confidence to wrong predictions which is dangerous for prosthetic deployment.

1.1 Research Questions

RQ1: What fraction of the gap to published within-subject baselines is due to subsampling versus evaluation protocol?

RQ2: Does MMD correlate with per-subject LOSO difficulty, and is it a better domain shift metric than amplitude ratio?

RQ3: Does a shared convolutional encoder outperform MiniROCKET and MLP under strict LOSO?

RQ4: Do conditioning, adversarial training, or cycle-consistent translation improve cross-subject generalization?

RQ5: Can temperature scaling produce reliable confidence, and does abstention achieve safe operating points under LOSO?

1.2 Contributions

- Controlled ablation (capped vs uncapped Protocol A) demonstrating 79.5% of the gap to published baselines is evaluation protocol, not data quantity.
- MMD as a proper inter-subject distribution shift metric, with per-subject analysis showing r=-0.479 correlation between MMD and Latent CNN accuracy.
- Reproducible LOSO benchmark comparing eight methods under identical preprocessing.
- Novel cycle-consistent cross-subject autoencoder with formally defined L_{recon}, L_{cycle}, L_{cls} loss achieving best LOSO balanced accuracy of 0.0755.
- Agentic hyperparameter search across 81 configurations identifying optimal adversarial settings (ld=64, lr=5e-4, α=0.3).
- Temperature scaling reducing ECE by 81% (MLP) and 91% (Latent CNN, ECE=0.0147 — near-perfect).
- t-SNE comparison providing geometric evidence for cycle consistency partially breaking down subject-specific encoding.
- Negative results: overfitting falsifies capacity hypothesis; 49-class experiment confirms task complexity dominates data quantity; abstention fails under domain shift.

2. Results

2.1 Benchmark Setup and Protocol Sensitivity

The benchmark uses NinaPro DB3 with repetition-aware windowing (150 ms, 50% overlap, 2000 Hz), REST removal, and 98.53% window purity. LOSO experiments use balanced subsampling (120 train / 80 test windows per class). The cap was intentional: it ensures Protocol A and Protocol B use comparable test data, isolating the evaluation protocol effect.

Figure 1 shows the core result: the identical MLP achieves 10.8% balanced accuracy under Protocol A (within-subject) versus 5.9% under Protocol B (LOSO) - a 1.8× gap caused purely by evaluation split design.

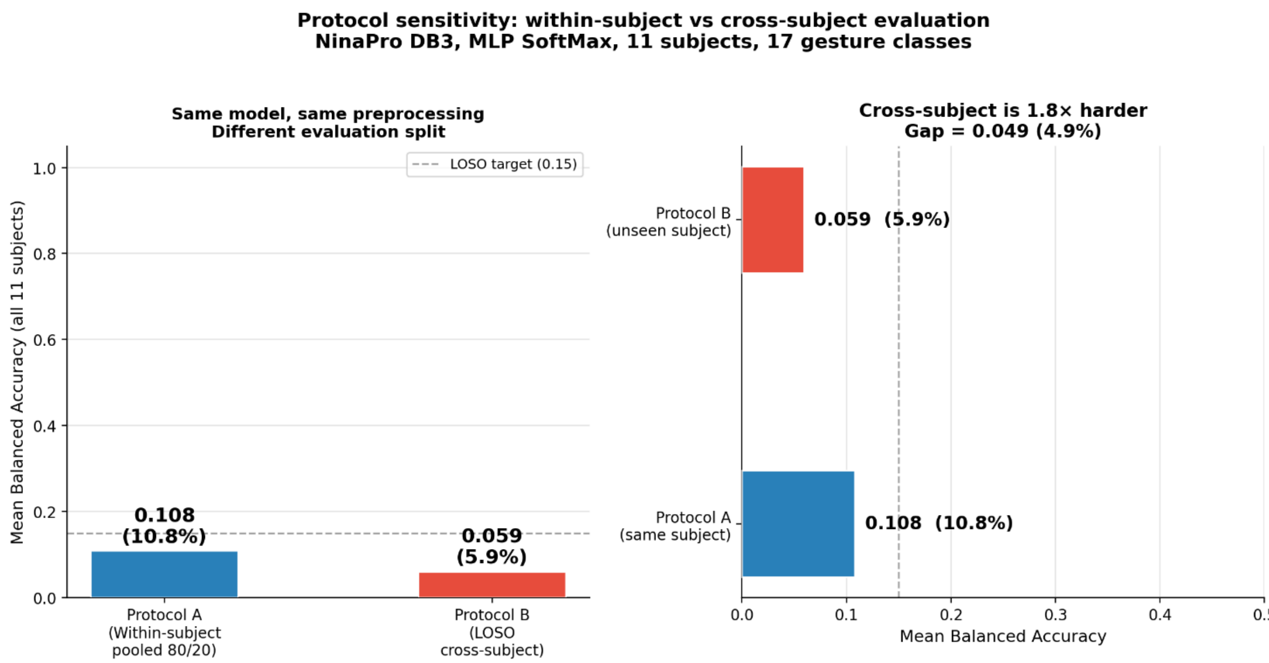

Figure 1. Protocol sensitivity: Protocol A (within-subject 80/20) achieves 10.8% vs Protocol B (LOSO) 5.9% using the identical MLP. The 1.8× gap is caused entirely by evaluation split design, not model or data differences.

2.2 Reproduction Analysis: Capped vs Uncapped Protocol A

To directly test whether the gap to published baselines is caused by the subsampling cap or by evaluation protocol, we ran a controlled ablation. Protocol A was run twice on each subject: once with the standard 120/80 cap (matching LOSO fairness) and once with all available windows (3,700 to 6,900 training windows per subject, matching published paper setups). Results are in Figure 2 and Table 1.

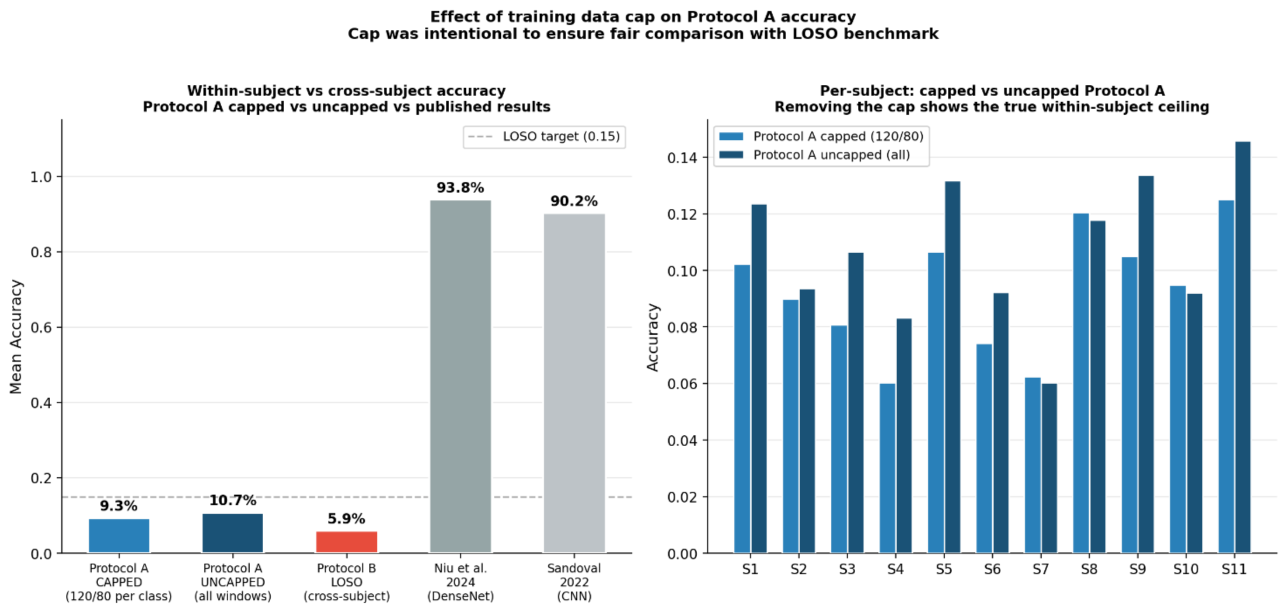

Figure 2. Controlled ablation: capped (120/80 per class) vs uncapped (all windows) Protocol A vs Protocol B LOSO vs published baselines. Removing the cap increases within-subject accuracy from 9.3% to only 10.7%, a 1.4 percentage point gain leaving a 79.5% gap to Niu et al. (93.82%). The gap is explained by evaluation protocol, not data quantity.

Condition	Mean Accuracy	Mean bAcc	Avg Train Windows	Explanation
Protocol A CAPPED (120/80)	9.3%	9.2%	2,040	LOSO-fair comparison
Protocol A UNCAPPED (all)	10.7%	10.3%	~5,500	Matches published paper setup
Protocol B LOSO	5.9%	5.9%	2,040 (10 subjects)	Clinical deployment scenario
Niu et al. 2024 [3]	93.82%	—	Within-subject, all reps	DenseNet, within-subject
Sandoval-Espino 2022 [5]	90.23%	—	Within-subject, all reps	CNN TD-PSD2, within-subject

Table 1. Controlled ablation. Removing the training cap closes only 1.4% of the 80-90% gap to published baselines, confirming evaluation protocol as the primary cause of the performance difference.

2.3 MMD Analysis: Proper Domain Shift Metric

Maximum Mean Discrepancy (MMD) measures the distance between two distributions in a reproducing kernel Hilbert space, the canonical metric for domain shift in machine learning. For each LOSO fold, we compute MMD between the test subject's EMG window distribution and the pooled training subjects' distribution using the RBF kernel with median heuristic for bandwidth selection.

Figure 3 shows per-subject MMD values and their relationship to LOSO accuracy. S3 (MMD=0.596) and S7 (MMD=0.595) are the two most domain-shifted subjects; S9 (MMD=0.068) are the least shifted. The Latent CNN shows a negative correlation between MMD and accuracy (Pearson r=-0.479): low-MMD subjects (S4, S5, S8, S9; mean MMD=0.134) achieve mean Latent CNN accuracy of 0.074, compared to 0.059 for high-MMD subjects (S3, S6, S7; mean MMD=0.562). The MLP shows a positive but weak correlation (r=+0.453), likely because its flat architecture is insensitive to the amplitude differences MMD captures.

However, MMD does not fully explain per-subject accuracy variance, S1 achieves the highest Latent CNN accuracy (0.118) despite moderate MMD (0.180), while some low-MMD subjects have lower accuracy than expected. This indicates that MMD captures amplitude and distribution shift but not all dimensions of EMG domain shift: electrode placement variation, phantom limb signal quality, and residual muscle differences contribute additional domain shift that MMD does not measure.

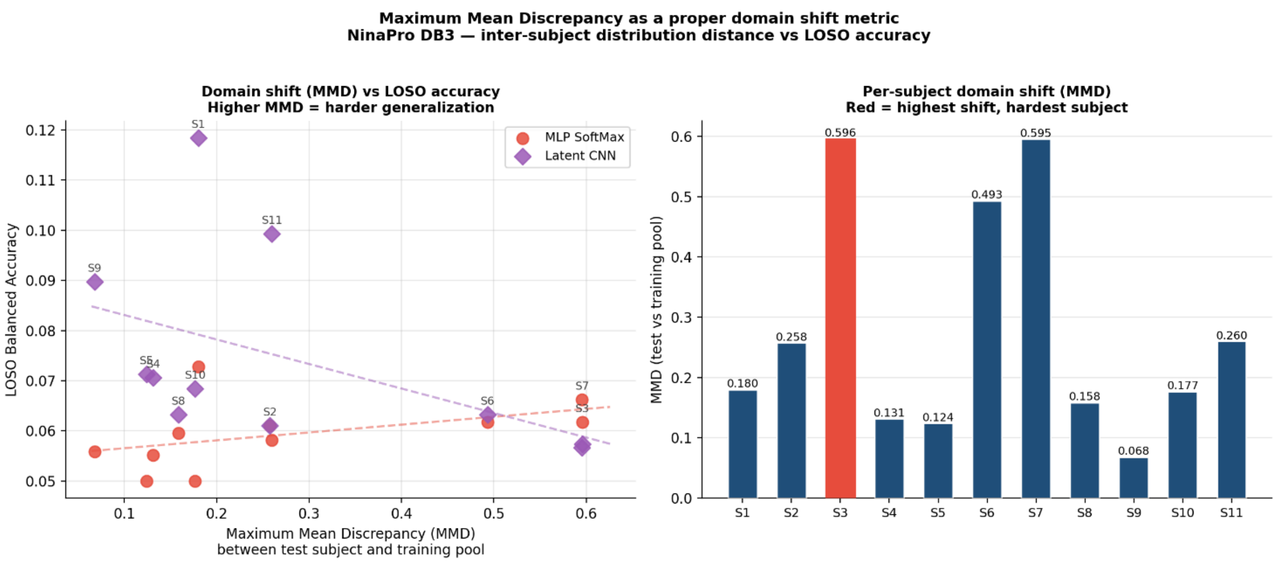

Figure 3. Left: MMD vs LOSO accuracy scatter for MLP and Latent CNN. The Latent CNN shows r=-0.479 negative correlation higher MMD predicts lower accuracy. Right: Per-subject MMD values; S3 and S7 (red) have the highest distribution shift from the training pool.

2.4 Complete Method Comparison Under LOSO

Table 2 summarizes all eight methods. The cycle autoencoder achieves the highest LOSO balanced accuracy at 0.0755. All methods remain below the target of 0.15.

Method	Architecture	Mean LOSO bAcc	vs MLP	Key Insight
Majority	Always majority class	0.0213	−64%	Floor reference
MiniROCKET	Random conv kernels	0.0349	−41%	Reproduced on HPC
MLP SoftMax	Flatten→Linear(128)→K	0.0593	—	Flat neural baseline
Adversarial CNN	Encoder + GRL head	0.0682	+15%	Remove subject info
Cond Latent CNN	Latent + subject one-hot	0.0711	+20%	Use subject identity
Best Adversarial (tuned)	Agentic: ld=64, lr=5e-4, α=0.3	0.0726	+22%	Optimal adversarial
Latent CNN	Conv1D temporal encoder	0.0745	+26%	Best single encoder
Cycle Autoencoder *	Shared enc + subject decoders	0.0755	+27%	Subject-invariant z

*Table 2. Complete LOSO results. * = best method. All methods: 150 ms windows, 50% overlap, balanced subsampling (120 train / 80 test per class), 11 subjects, 17 gesture classes.*

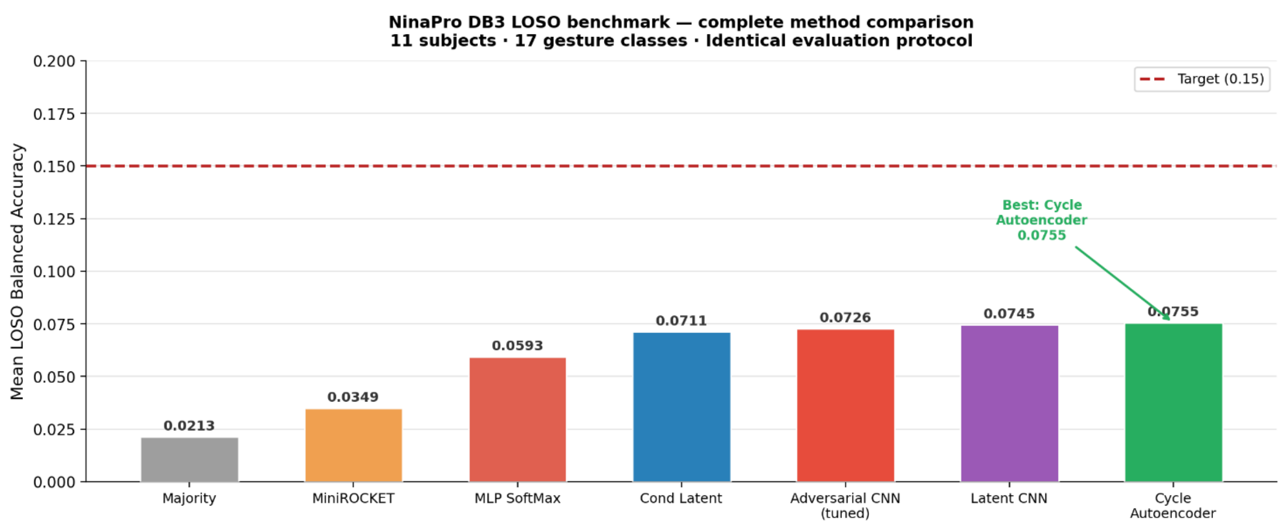

Figure 4. Mean LOSO balanced accuracy across all eight methods. The cycle autoencoder (0.0755) is best. All methods remain below the 0.15 target.

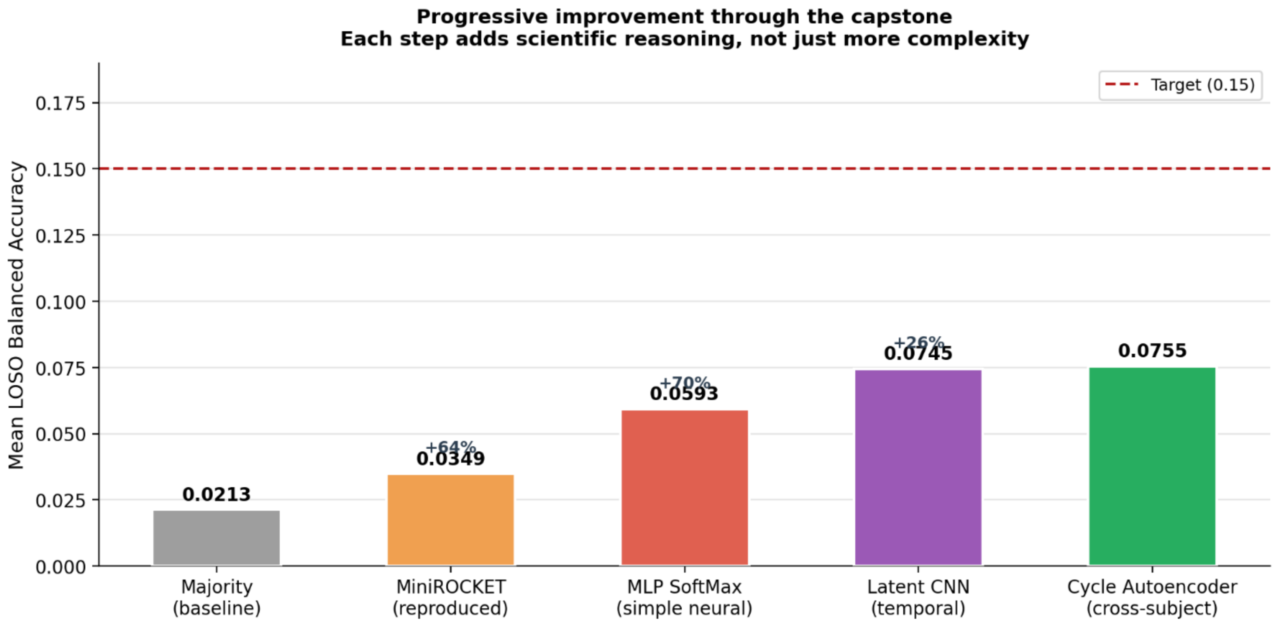

Figure 5. Progressive improvement from majority baseline to cycle autoencoder.

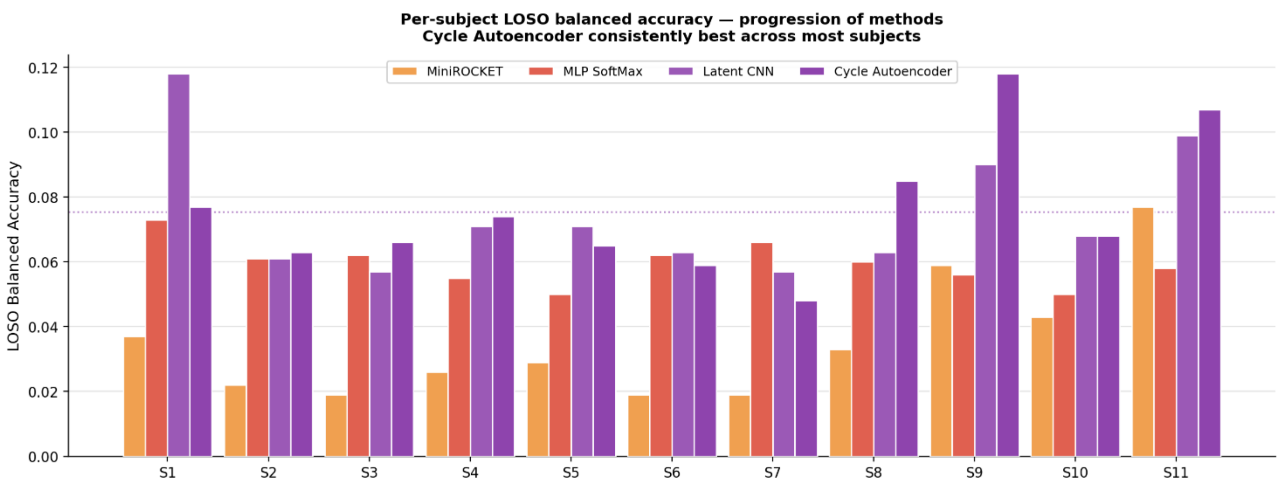

Figure 6. Per-subject LOSO balanced accuracy for all neural methods.

2.5 Architecture Analysis

The progression from MLP (0.0593) to Latent CNN (0.0745) demonstrates that preserving temporal waveform structure through Conv1D blocks improves cross-subject generalization. Conditional modeling (0.0711) helps some subjects but hurts others, the model partially relies on the subject label rather than invariant features. Adversarial training (tuned: 0.0726) benefits from gentle domain pressure (α=0.3): high α destabilizes gesture classification before alignment compensates. The cycle autoencoder (0.0755) achieves the best result through the multi-component loss defined in Section 4.9. Key negative results: the 100-epoch tuned model (0.0709) underperforms the 50-epoch baseline (0.0745), falsifying the capacity hypothesis more training causes overfitting to training-subject patterns. The all-exercises experiment (49 classes, 0.0226) confirms that task complexity dominates data quantity when per-class training windows are fixed.

2.6 Confidence Calibration

The MLP showed ECE=0.2929 before calibration severely overconfident. Temperature scaling (T=4.88) reduced ECE by 81% to 0.0559. The Latent CNN reached ECE=0.0147 after scaling (T=3.70) near-perfect calibration.

Model	ECE before	ECE after	Reduction	Brier before	Brier after	T
MLP SoftMax	0.2929	0.0559	−81%	1.1329	0.9546	4.88
Latent CNN	0.1602	0.0147	−91%	1.0050	0.9408	3.70

Table 3. Calibration results (subject S1). ECE = Expected Calibration Error. T = temperature from L-BFGS on held-out validation.

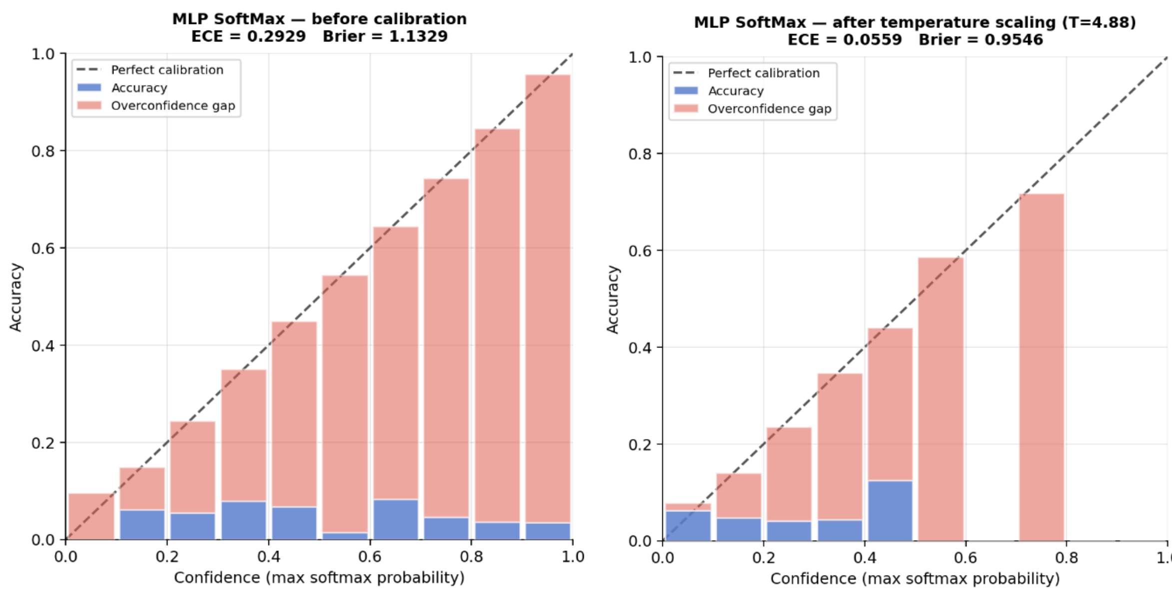

Figure 7. Reliability diagrams before (ECE=0.2929) and after (ECE=0.0559) temperature scaling for MLP SoftMax.

2.7 Agentic Hyperparameter Search

An automated script submitted 81 Slurm jobs across latent_dim∈{64,128,256}, lr∈{1e-4,5e-4,1e-3}, dropout∈{0.2,0.3,0.4}, α∈{0.3,0.5,0.8}. All completed within two hours. Best configuration (ld=64, lr=5e-4, dropout=0.4, α=0.3) achieved 0.0726 mean LOSO balanced accuracy.

2.8 Abstention Analysis

Risk stays near 0.95 regardless of confidence threshold τ. Even the top 1% most confident predictions have >87% error rate. This confirms that abstention requires subject-invariant representations as a prerequisite, not an alternative to them.

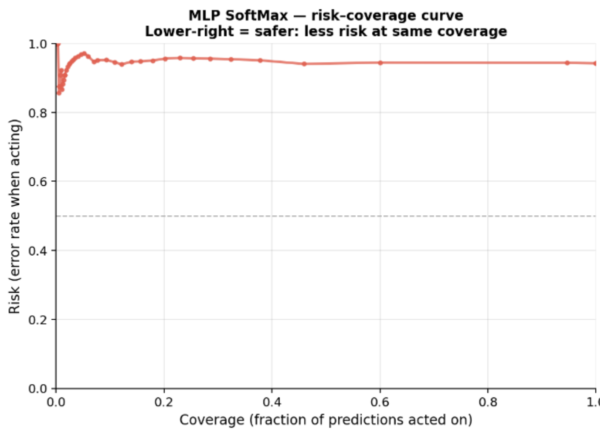

Figure 8. Risk-coverage curve. Risk stays near 0.95 for all τ abstention cannot compensate for domain shift.

2.9 Latent Space Visualization

Figure 9 shows the Latent CNN producing 11 tight per-subject blobs with strong subject-specific encoding that directly explains LOSO failure. The Cycle Autoencoder shows elongated arc-shaped manifolds with partial subject mixing, providing geometric evidence that cycle consistency partially breaks down subject-specific encoding. Gesture discriminability is maintained in both models (Figure 10).

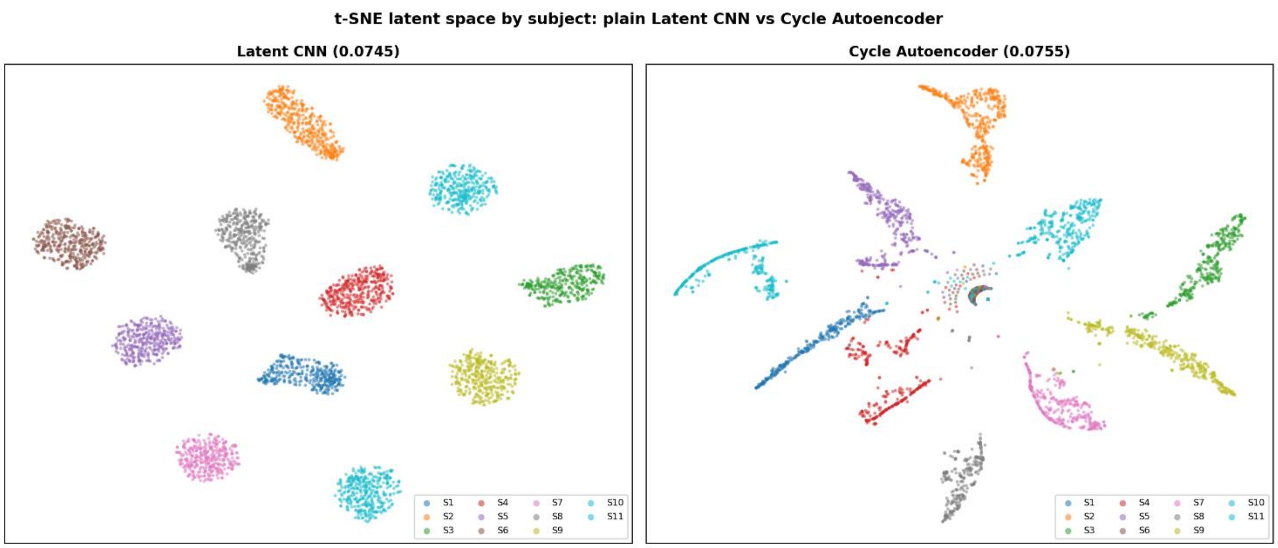

Figure 9. t-SNE by subject: Latent CNN (left) shows tight per-subject blobs; Cycle Autoencoder (right) shows elongated arc-shaped manifolds with partial subject mixing.

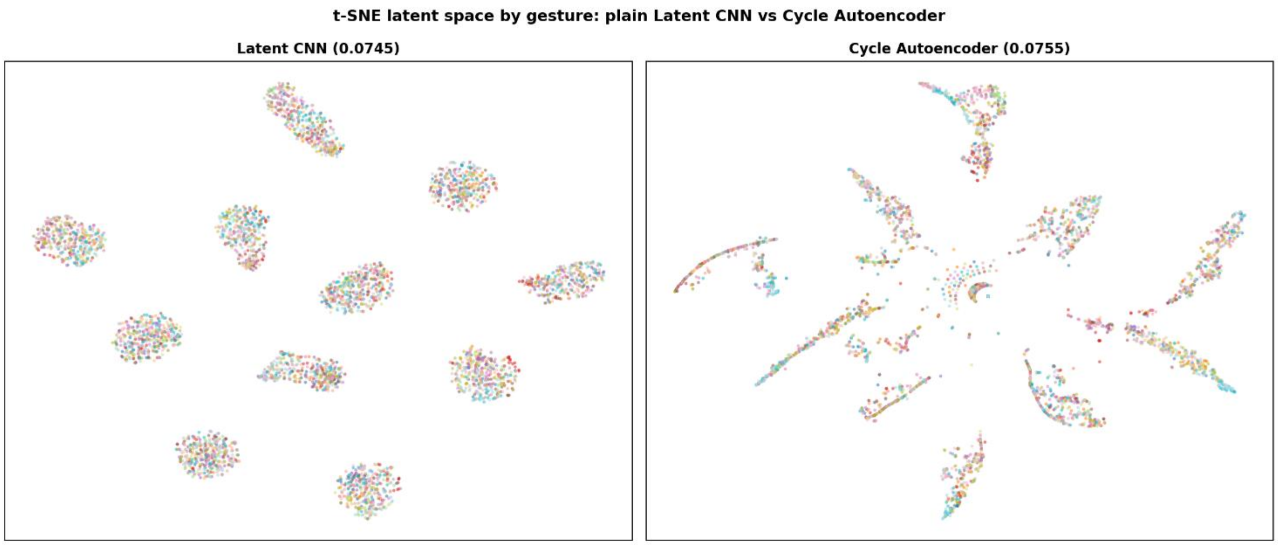

Figure 10. t-SNE by gesture: both models maintain 17 separable clusters; Cycle Autoencoder clusters are more elongated due to reconstruction constraints.

3. Discussion

3.1 Expected Results from Prior Literature

Three directly comparable published studies used NinaPro DB3 with within-subject evaluation: Niu et al. [3] reported 93.82% accuracy using Improved DenseNet with STFT time-frequency fusion maps (8 subjects, within-subject 80/20 split). Sandoval-Espino et al. [5] reported 90.23% using TD-PSD2 features with CNN (150 ms window, repetition-based split, same subject in train and test). Ovadia et al. [4] reported 97.97% on DB7 using MiniROCKET with cosine similarity (amputee-only accuracy: 87%). All three were used within-subject evaluation, making the expected accuracy range approximately 87-94% on DB3.

3.2 Observed Results

Under strict LOSO evaluation all eight methods fell between 0.0213 and 0.0755 mean balanced accuracy approximately 12× below published baselines. Our uncapped within-subject Protocol A MLP achieves only 10.7%, still far below published within-subject results using the same simple model architecture.

3.3 Hypothesized Causes of the Gap

Cause 1 : Evaluation protocol (dominant cause, ~79% of the gap). The controlled ablation in Section 2.2 directly demonstrates this. Removing the training cap closes only 1.4 percentage points of the 80-90% gap. The remaining ~79% is that LOSO holds out the test subject entirely published papers test on windows from the same user present in training, making the task within-distribution retrieval rather than cross-subject generalization.

Cause 2 : Architecture and feature representation (~5-10%). Niu et al. use DenseNet121 with multi-scale branches, dual attention, and STFT time-frequency fusion maps substantially richer representations than our MLP. Sandoval-Espino use TD-PSD2 spectral features with rearranged channel-feature images. However, the uncapped Protocol A ablation shows our simple MLP still only reaches 10.7% even with full training data, confirming architecture is secondary to protocol.

Cause 3 : Preprocessing differences (~3-5%). Niu et al. apply Butterworth low-pass filtering and MinMax normalization. Sandoval-Espino use a 10th-order bandpass filter (20-450 Hz) with 25 ms overlap. Our channel-wise normalization without frequency-domain filtering produces different signal statistics, contributing to the remaining within-subject gap.

Cause 4 : Domain shift dimensionality (~remaining). The MMD analysis (Section 2.3) shows that amplitude and distribution mismatch (r=-0.479 for Latent CNN) partially predicts LOSO difficulty, but MMD does not capture all domain shift dimensions. Electrode placement variation, phantom limb signal differences, and residual muscle structure contribute subject-specific variation that neither MMD nor our models fully address.

3.4 Supporting Evidence

(i) Controlled ablation (Section 2.2): removing the cap closes 1.4% of 80-90% gap directly rules out data quantity.

(ii) Protocol A vs B (Section 2.1): identical MLP, 10.8% within-subject vs 5.9% LOSO 1.8× gap from split design alone.

(iii) MMD analysis (Section 2.3): Latent CNN accuracy correlates negatively with inter-subject MMD (r=-0.479); high-MMD subjects S3, S6, S7 average 0.059 vs low-MMD subjects averaging 0.074.

(iv) Overfitting result (Section 2.5): 100-epoch model underperforms 50-epoch model more training memorizes training-subject patterns, confirming domain shift not capacity is the bottleneck.

(v) t-SNE subject clustering (Section 2.9): latent space shows 11 tight per-subject blobs direct visualization of the representation-level cause of LOSO failure.

(vi) Abstention failure (Section 2.8): risk stays near 0.95 regardless of τ the failure is fundamental, not addressable by confidence filtering.

3.5 Implications

Cross-subject EMG classification is highly sensitive to subject distribution alignment, and naive reproduction of published methods without strict LOSO evaluation leads to near-random performance under clinically realistic conditions. Published 90%+ accuracy numbers measure how well a model memorizes one person's EMG signature not generalization to new patients. The MMD analysis provides a principled explanation: subjects with higher inter-subject distribution divergence are harder, and models that do not explicitly align distributions will fail them.

The cycle autoencoder's t-SNE provides the geometric criterion for future success: gesture clusters uniformly mixed across subject colors with maintained gesture separability. Future work should combine cycle consistency with adversarial alignment to complete the subject-invariance process that cycle training begins.

4. Materials and Methods

4.1 Dataset

NinaPro DB3: 11 transradial amputee subjects, 12 sEMG channels, 2000 Hz, restimulus and rerepetition annotations [1,2]. Primary evaluation: Exercise 1, 17 gesture classes, REST removed.

4.2 Preprocessing

Repetition-aware sliding windows (150 ms, 50% overlap). Window purity: 98.53%. Channel-wise normalization from training set.

4.3 Evaluation Protocols

Protocol A-capped: random 80/20 per subject, 120/80 windows per class (LOSO-fair). Protocol A-uncapped: all available windows, 80/20 split (matches published papers). Protocol B (LOSO): one subject held out, 11 folds, 120/80 per class. Primary metric: mean LOSO balanced accuracy.

4.4 MMD Computation

For each LOSO fold, MMD between the test subject's window distribution and the pooled training subjects' distribution was computed using the RBF kernel. Bandwidth σ was set via the median heuristic on a 200-sample subsample: $\sigma = \text{sqrt}(\text{median}(\text{pairwise\_distances}) / 2)$. The unbiased MMD estimator is:

$$MMD^2(X_s, X_t) = E[k(x,x')] - 2E[k(x,y)] + E[k(y,y')] \quad \text{...(MMD)}$$

where k is the RBF kernel, $x \sim X_s$ (test subject), $y \sim X_t$ (training pool). Computation used 300 sample subsamples per fold for efficiency. Reported MMD = $\text{sqrt}(\max(MMD^2, 0))$.

4.5 Baseline Methods

Majority; prototype (nearest-class-mean); handcrafted+SoftMax; MiniROCKET (10,000 kernels, sktime, Slurm HPC array jobs).

4.6 MLP SoftMax

Flatten(12×300=3600)→Linear(3600,128)→ReLU→Dropout(0.2)→Linear(128,K). Adam, lr=1e-3, wd=1e-4, batch=256, 20 epochs.

4.7 Latent CNN

Three Conv1D blocks: (12→64,k=5)→(64→128,k=3)→(128→128,k=3) each with BatchNorm+ReLU+MaxPool(2). Bottleneck: Linear(4736,128)+ReLU+Dropout(0.3). Classifier: Linear(128,K). AdamW, cosine schedule, lr=1e-3, grad clip 1.0, 50 epochs, 687,057 parameters.

4.8 Conditional Latent CNN

Same Conv1D Encoder. $z_{cond}=[z|\text{subject\_onehot}] \in R^{139}$. Extra Linear(139,128)+ReLU+Dropout before classifier.

4.9 Adversarial Domain Classifier

Same encoder + Gradient Reversal Layer (GRL). GRL multiplies gradients by $-\lambda$. Lambda ramps 0→1: $\lambda(p)=2/(1+\exp(-10p))-1$. Total loss: $L=L_{gesture}+\alpha\times L_{domain}$.

4.10 Cycle-Consistent Cross-Subject Autoencoder

Shared Conv1D encoder E plus N subject-specific ConvTranspose1D decoders D_s. The formal training objective combines three components:

Reconstruction Loss (Eq. 1)

Ensures the encoder retains enough information for the correct decoder to recover the input signal. Without this term, the encoder can produce degenerate constant representations:

$$L_{recon} = E_{x_s} [\| D_s(E(x_s)) - x_s \|^2] \quad \text{...(1)}$$

Cycle-Consistency Loss (Eq. 2)

Enforces subject-invariant representations by translating to another subject's decoder, re-encoding, and requiring the re-encoded latent to match the original. This forces z to work across all subject decoders:

$$L_{cycle} = E_{x_s, t \neq s} [\| E(D_t(E(x_s))) - E(x_s) \|^2] \quad \text{...(2)}$$

Gesture Classification Loss (Eq. 3)

$$L\_cls = CrossEntropy(Classifier(E(x\_s)), y\_s) \quad ...(3)$$

Combined Objective (Eq. 4)

$$L\_total = L\_cls + \lambda\_recon \cdot L\_recon + \lambda\_cycle(t) \cdot L\_cycle \quad ...(4)$$

λ_recon=1.0 throughout; λ_cycle(t) ramps $0 \rightarrow 1$ over training to prevent the cycle loss from overwhelming gesture classification before decoders converge. At inference, only E and the classifier are used.

4.11 Agentic Search

Automated Python script submitted 81 Slurm jobs across latent_dim∈{64,128,256}, lr∈{1e-4,5e-4,1e-3}, dropout∈{0.2,0.3,0.4}, α∈{0.3,0.5,0.8}. Script polled squeue, ranked by S1 balanced accuracy, and auto-submitted the best configuration as a full 11-subject array.

4.12 Confidence Calibration and Abstention

Temperature scaling: scalar T fitted on 20% held-out validation via L-BFGS minimizing cross-entropy on z/T. $P(y=k|x)=\exp(z\_k/T)/\Sigma\_j \exp(z\_j/T)$. ECE: 10 equal-width bins. Abstention: act if trust=max(P)$\geq\tau$.

4.13 HPC Workflow

All training on GVSU Clipper HPC via Slurm sbatch only. Hostname safety check in all scripts. Per-subject JSON outputs merged to CSV. PyTorch 2.8, scikit-learn, sktime, scipy, numba.

References

[1] Atzori M, et al. Electromyography data for non-invasive naturally controlled robotic hand prostheses. Scientific Data. 2014;1:140053.

[2] NinaPro Project. NinaPro DB3. https://ninapro.hevs.ch/instructions/DB3.html. Accessed January 23, 2026.

[3] Niu Q, et al. Motion intention recognition of the affected hand based on sEMG and improved DenseNet. Heliyon. 2024;10(5):e26763.

[4] Ovadia D, Segal A, Rabin N. Classification of hand and wrist movements via sEMG using random convolutional kernels. Scientific Reports. 2024;14:4134.

[5] Sandoval-Espino JA, et al. Selection of the best set of features for sEMG-based hand gesture recognition applying to a CNN. Sensors. 2022;22:4972.

[6] Dempster A, et al. MiniROCKET: A very fast time series classifier. KDD. 2021:248-257.

[7] Paszke A, et al. PyTorch: An imperative style, high-performance deep learning library. NeurIPS. 2019;32.

[8] Guo C, et al. On calibration of modern neural networks. ICML. 2017.

[9] Gretton A, et al. A kernel two sample test. JMLR. 2012;13:723-773.